

Multiple Choice (10 marks)

1. What endogenous substrate source provides the most energy during moderate to high intensity exercise?
 - (a) Liver glycogen
 - (b) Muscle glycogen
 - (c) Intramuscular lipid
 - (d) Adipose tissue lipid
2. Which of the following foods do not contain free sugars
 - (a) Cows' milk
 - (b) Natural unsweetened orange juice
 - (c) Honey
 - (d) Agave nectar
3. Which of the following vitamins is required for fatty acid β -oxidation?
 - (a) Niacin
 - (b) Riboflavin
 - (c) Thiamin
 - (d) Vitamin C
4. Which of the following statements about glucose metabolism is correct?
 - (a) Glucagon increases the rate of glycolysis
 - (b) Glycolysis requires NADP^+
 - (c) In glycolysis, glucose is cleaved into two three-carbon products
 - (d) The end-product of glycolysis in red blood cells is pyruvate
5. Which of the following is a function of folate?
 - (a) Synthesis of TMP (thymidine monophosphate)
 - (b) Decarboxylation of amino acids to form amine neurotransmitters
 - (c) Synthesis of methionine from homocysteine
 - (d) Carboxylation of pyruvate to oxaloacetate

True / False (10 marks)

Answer True or False

6. True or False: The fear of becoming fat in patients with anorexia nervosa is classified as a paranoid delusion, indicating a severe misinterpretation of reality.
7. True or False: A raw potato has a lower glycaemic index compared to a baked apple or apple juice.
8. True or False: Older adults typically gain muscle mass as they age due to increased physical activity.
9. True or False: Muscle contains less water than fat, which slows down the travel of electrical currents.
10. True or False: Potassium is a dietary component that has been shown to help reduce hypertension in humans.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how dietary factors, particularly salt and salt-preserved foods, may influence the risk of developing stomach cancer. Include explanations of the mechanisms involved.
12. Discuss the impact of in utero exposure to the Dutch Hunger Winter on DNA methylation in blood cells, focusing specifically on the significance of the first trimester of pregnancy.

End of Paper